

Assignment 4: Markov Chain Monte Carlo (MCMC)

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- Course: Machine Learning 2
- Prof: Jan Nagler
- Due: 17 May 2026

The devised program estimates robustly, given very noisy and very sparse data of infected and recovered of a past epidemic, the basic reproduction number of the SIR model. To keep computation in limit, we assume $\gamma=1$. The SIR model is implemented in a minimal and optimal way, using scaled variables and a scaled time. Only the ODE part is numerically integrated that needs to be integrated. The noisy number of infected and the number of recovered are highly correlated. This relationship helps MCMC infer the parameters.

Get familiar with the commented MCMC code below.

Task: Change the program to the SIRD model, by including (D)eaths, with rate μ . Fix not only $\gamma = 1$ but also $\beta = 2.5$ (or to a higher value of your choice). Infer the death rate μ , given noisy $S(t)$, $I(t)$, $R(t)$, $D(t)$ input curves. If you want, you can try to optimize the code (optional, very very hard). Also optional is: Does the inference for μ work, if $S(t)$ and/or $R(t)$ are not given? You may use these (initial) conditions/parameters:

$$i0 = 0.01, s0 = 0.99, r0 = 0, d0 = 0, f = 3.0, timestep = 0.5.$$

You may assume values for the respective σ 's (log-normal noises) in the range of 0.2 – 0.4, but not lower than 0.1. For assessment use HPD plots, or similar. Good luck and have fun.

Step 1: Import and Settings

```
In [10]: # macOS / PyTensor compiler workaround. Run this BEFORE importing pymc.
# It fixes cases where clang++ cannot find the C++ standard library header <vector>.
import os
import sys
import subprocess

if sys.platform == "darwin":
    try:
        sdk = subprocess.check_output(["xcrun", "--show-sdk-path"], text=True).strip()
        os.environ.setdefault("SDKROOT", sdk)
        import pytensor
        pytensor.config.cxx = "/usr/bin/clang++"
        pytensor.config.gcc__cxxflags = f"-I{sdk}/usr/include/c++/v1"
        print("Configured PyTensor C++ flags for macOS SDK:", pytensor.config.gcc__cxxflags)
```

```
except Exception as exc:
    print("Could not configure PyTensor compiler workaround:", repr(exc))
```

Configured PyTensor C++ flags for macOS SDK: -I/Library/Developer/CommandLineTools/SDKs/MacOSX.sdk/usr/include/c++/v1

```
In [2]: # Assignment 4: SIRD model, MCMC for mu
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
sns.set_theme()
import pymc as pm #install if necessary
from pymc.ode import DifferentialEquation
from scipy.integrate import odeint
import warnings
warnings.filterwarnings("ignore")
from warnings import simplefilter
# ignore all future warnings
simplefilter(action='ignore', category=FutureWarning)
np.random.seed(42)
```

/Applications/anaconda3/envs/DataAnalytics/lib/python3.13/site-packages/arviz/__init__.py:50: FutureWarning: ArviZ is undergoing a major refactor to improve flexibility and extensibility while maintaining a user-friendly interface. Some upcoming changes may be backward incompatible. For details and migration guidance, visit: https://python.arviz.org/en/latest/user_guide/migration_guide.html

warn(

Step 2: Defining SIRD ODE

The SIRD model extends the classic SIR model by adding a **Deaths (D)** compartment. The system of ODEs is:

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I - \mu I$$

$$\frac{dR}{dt} = \gamma I$$

$$\frac{dD}{dt} = \mu I$$

Parameter	Value	Role
β	2.5 (fixed)	Transmission rate
γ	1.0 (fixed)	Recovery rate

Parameter	Value	Role
μ	inferred	Death/mortality rate

Initial conditions: $s_0 = 0.99$, $i_0 = 0.01$, $r_0 = 0$, $d_0 = 0$

```
In [9]: # Define initial conditions of SIRD model
s0=0.99
i0 = 0.01 #fractions infected at time t0=0 (1%)
r0 = 0.00 #fraction of recovered at time t0=0
d0=0.00 #fraction of recovered at time t0=0
f = 3 #3.0 # time factor, defines total time window range
timestep_data = 0.5 # dt for data (e.g., weekly)

# ODE SIRD system: dS/dt = -beta*S*I, dI/dt = beta*S*I - gamma*I - mu*I, dR/dt = gamma*I, dD/dt = mu*I
# beta = 2.5, gamma = 1.0 are fixed, mu is inferred (p[0])
def SIRD(y, t, p):
    s = y[0] # Susceptible
    i = y[1] # Infected
    r = y[2] # Recovered
    d = y[3] # Dead

    mu_infer = p[0] # The parameter to be inferred is mu
    beta_fixed = 2.5 # Fixed beta as per task
    gamma_fixed = 1.0 # Fixed gamma as per task

    ds_dt = -beta_fixed * s * i
    di_dt = beta_fixed * s * i - gamma_fixed * i - mu_infer * i
    dr_dt = gamma_fixed * i
    dd_dt = mu_infer * i
    return [ds_dt, di_dt, dr_dt, dd_dt] # Return derivatives in order of states: S, I, R, D

times = np.arange(0,5*f,timestep_data)

#ground truth (fixed gamma=1, then R0=beta, time scale to t/gamma)
# This variable is not used in SIRD function, but used for reference
```

Step 3: Generating synthetic “true” epidemic curves and add noise

We generate ground-truth SIRD curves using `odeint` with the true death rate $\mu_{\text{true}} = 0.15$, then corrupt each compartment with additive Gaussian noise ($\sigma = 0.2$) to simulate realistic, noisy observations.

This lets us verify later whether MCMC recovers the true parameter.

```
In [11]: # ODE system container is now defined within the pm.Model context in cell s8QFuLNuKeLT
```

```
mu_true = 0.15

times = np.arange(0, 5*f, timestep_data)
#create SIRD Curves
solution = odeint(
    SIRD,
    [s0, i0, r0, d0],
    times,
    args=(mu_true,)
)

S_data = solution[:,0]
I_data = solution[:,1]
D_data = solution[:,3]
R_data = solution[:,2]
noise = 0.2

S_obs = S_data + np.random.normal(0, noise, len(S_data))
I_obs = I_data + np.random.normal(0, noise, len(I_data))
R_obs = R_data + np.random.normal(0, noise, len(R_data))
D_obs = D_data + np.random.normal(0, noise, len(D_data))
```

Step 4: Building PyMC model

We use PyMC's `DifferentialEquation` wrapper to embed the SIRD ODE inside a probabilistic model.

Priors:

- $\mu \sim \text{LogNormal}(\log(0.1), 0.5)$ — enforces $\mu > 0$, broad enough to not dominate
- $\sigma \sim \text{HalfCauchy}(1)$ — weakly informative noise prior per compartment

Likelihood: Each observed compartment is modeled as `Normal(ODE_prediction,  $\sigma$ )`.

Sampling settings: 2000 draws, 1000 tuning steps, 4 chains, target_accept = 0.9

```
In [12]: with pm.Model() as model:
    # ODE system container
    sird_model = DifferentialEquation(
        func = SIRD,
        times = np.arange(0,5*f,timestep_data), # Reverted to plain numpy array for times
        n_states = 4, #r(ecovered) and i(nfected) are states
        n_theta = 1, # beta=R0 only parameter
        t0 = 0 # start from zero
```

```

)

mu = pm.Lognormal("mu", mu=np.log(0.1), sigma=0.5)
sigma = pm.HalfCauchy("sigma", 1, shape=4)

sird_curves = sird_model(
    y0=np.array([s0, i0, r0, d0]), # Changed y0 to a NumPy array for better shape inference
    theta=[mu]
)

S_hat = sird_curves[:, 0]
I_hat = sird_curves[:, 1]
R_hat = sird_curves[:, 2]
D_hat = sird_curves[:, 3]

S_like = pm.Normal(
    "S_like",
    mu=S_hat,
    sigma=sigma[0],
    observed=S_obs
)

I_like = pm.Normal(
    "I_like",
    mu=I_hat,
    sigma=sigma[1],
    observed=I_obs
)

R_like = pm.Normal(
    "R_like",
    mu=R_hat,
    sigma=sigma[2],
    observed=R_obs
)

D_like = pm.Normal(
    "D_like",
    mu=D_hat,
    sigma=sigma[3],
    observed=D_obs
)

```

```

In [8]: with model:
    trace = pm.sample(
        draws=2000,
        tune=1000,
        chains=4,

```

```

        target_accept=0.9,
        random_seed=42,
        return_inferencedata=True
    )
print(pm.summary(trace, var_names=["mu"]).round(2))

```

Initializing NUTS using jitter+adapt_diag...

Multiprocess sampling (4 chains in 4 jobs)

NUTS: [mu, sigma]

Output()

Sampling 4 chains for 1_000 tune and 2_000 draw iterations (4_000 + 8_000 draws total) took 407 seconds.

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
mu	0.12	0.03	0.07	0.18	0.0	0.0	8040.0	4784.0	1.0

Step 5: Displaying model plots

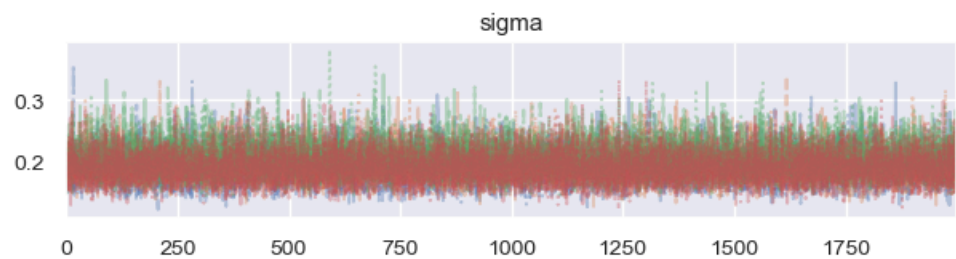
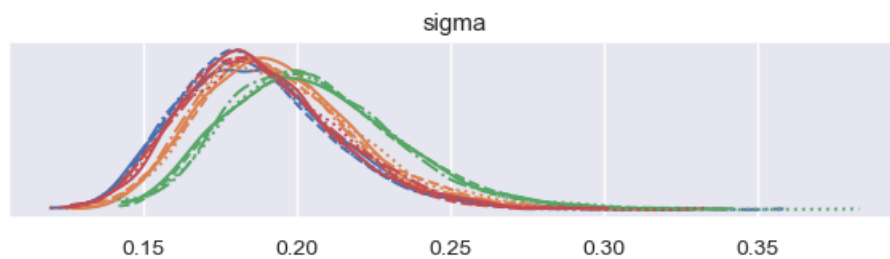
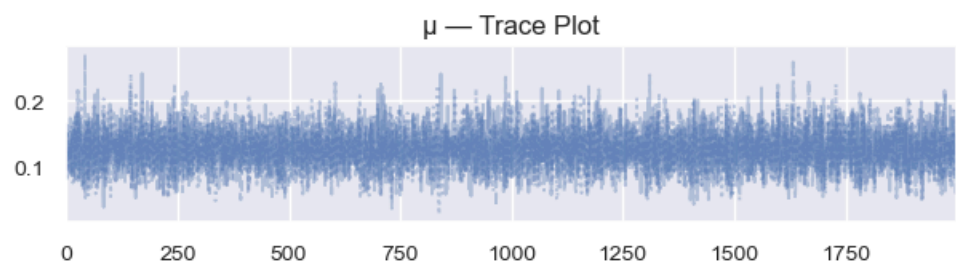
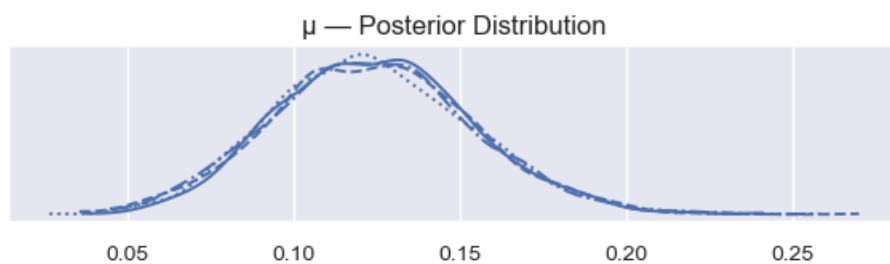
In [14]: `import arviz as az`

```

# Trace plot
axes = az.plot_trace(trace)
axes[0, 0].set_title("μ — Posterior Distribution") # top-left (KDE)
axes[0, 1].set_title("μ — Trace Plot")             # top-right (trace)
plt.suptitle("S, I, R, D observed", fontsize=14, y=1.02)
plt.tight_layout()
plt.show()

```

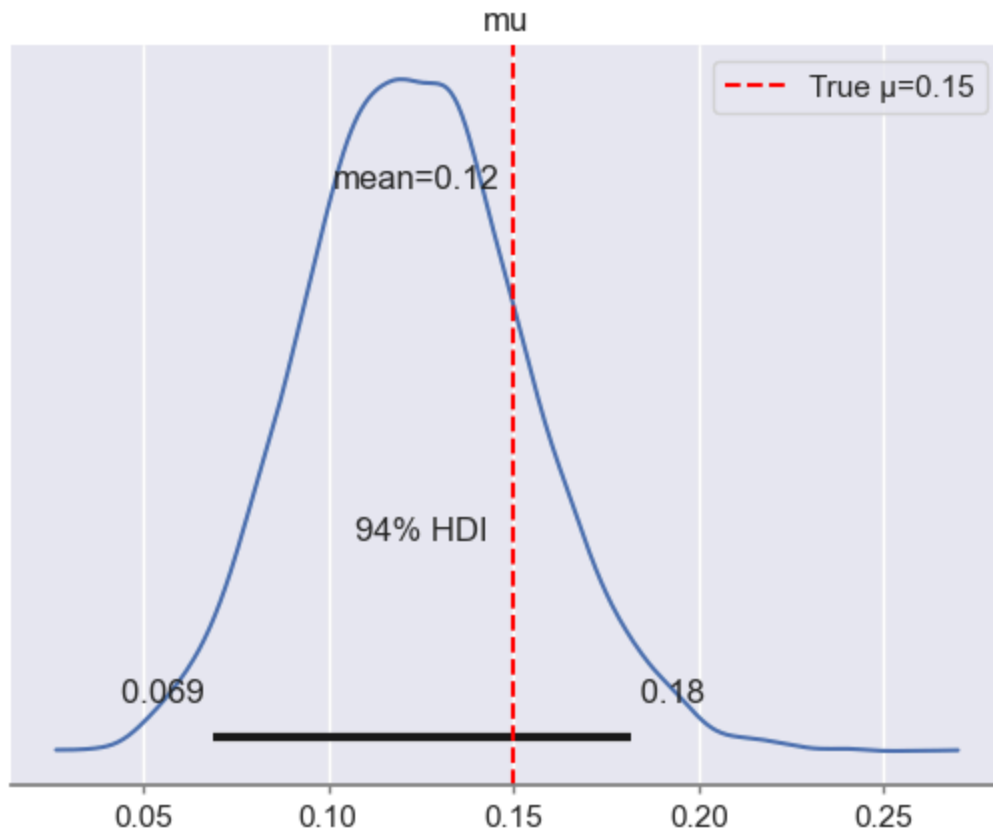
S, I, R, D observed



Posterior plot for μ with true value reference

```
In [ ]: az.plot_posterior(trace, var_names=["mu"], hdi_prob=0.94)
plt.axvline(mu_true, color='red', linestyle='--', label='True  $\mu=0.15$ ')
plt.legend()
```

Out[]: <matplotlib.legend.Legend at 0x17356d6a0>



If you want: Further Reading (see slides!), also: https://www.pymc.io/projects/examples/en/latest/diagnostics_and_criticism/Diagnosing_biased_Inference_with_Divergences.html

The following three sections address the optional question:

Does inference for μ work if $S(t)$ and/or $R(t)$ are not given?

We test three reduced models:

Model	Observed Compartments
model1	I, R, D (no S)
model2	S, I, D (no R)
model3	I, D only

Step 6: Reduced model - no S observed

```
In [16]: with pm.Model() as model1:
# ODE system container
sird_model = DifferentialEquation(
    func = SIRD,
    times = np.arange(0,5*f,timestep_data), # Reverted to plain numpy array for times
    n_states = 4, #r(ecovered) and i(nfected) are states
    n_theta = 1, # beta=R0 only parameter
    t0 = 0 # start from zero
)

mu = pm.Lognormal("mu", mu=np.log(0.1), sigma=0.5)
sigma = pm.HalfCauchy("sigma", 1, shape=3)

sird_curves = sird_model(
    y0=np.array([s0, i0, r0, d0]), # Changed y0 to a NumPy array for better shape inference
    theta=[mu]
)
#S(t) value is considered here as not known
I_hat = sird_curves[:, 1]
R_hat = sird_curves[:, 2]
D_hat = sird_curves[:, 3]

I_like = pm.Normal(
    "I_like",
    mu=I_hat,
    sigma=sigma[0],
    observed=I_obs
)
```



```

R_like = pm.Normal(
    "R_like",
    mu=R_hat,
    sigma=sigma[1],
    observed=R_obs
)
D_like = pm.Normal(
    "D_like",
    mu=D_hat,
    sigma=sigma[2],
    observed=D_obs
)

```

```

In [17]: with model1:
    trace1 = pm.sample(
        draws=2000,
        tune=1000,
        chains=3,
        target_accept=0.9,
        random_seed=42,
        return_inferencedata=True
    )
print(pm.summary(trace1, var_names=["mu"]).round(2))

```

Initializing NUTS using jitter+adapt_diag...

Multiprocess sampling (3 chains in 3 jobs)

NUTS: [mu, sigma]

Output()

Sampling 3 chains for 1_000 tune and 2_000 draw iterations (3_000 + 6_000 draws total) took 328 seconds.

We recommend running at least 4 chains for robust computation of convergence diagnostics

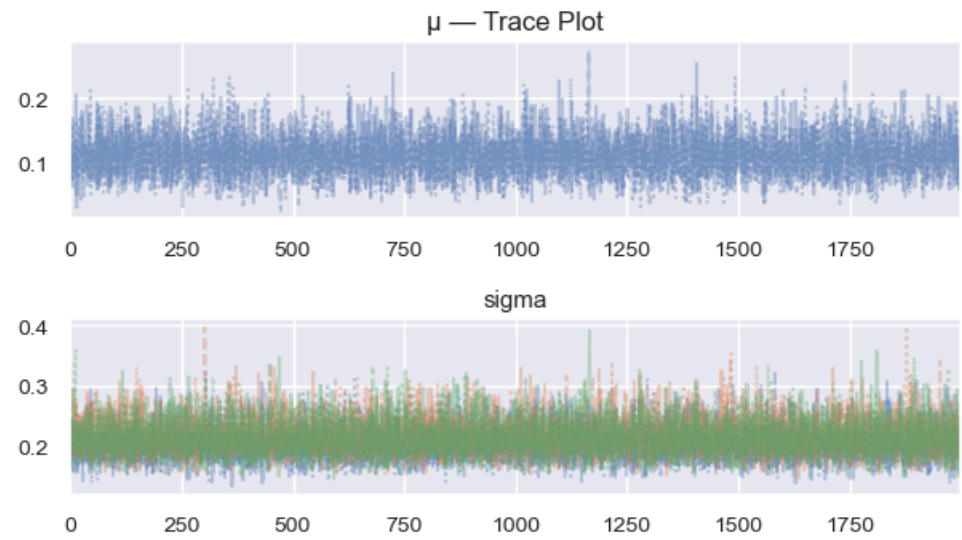
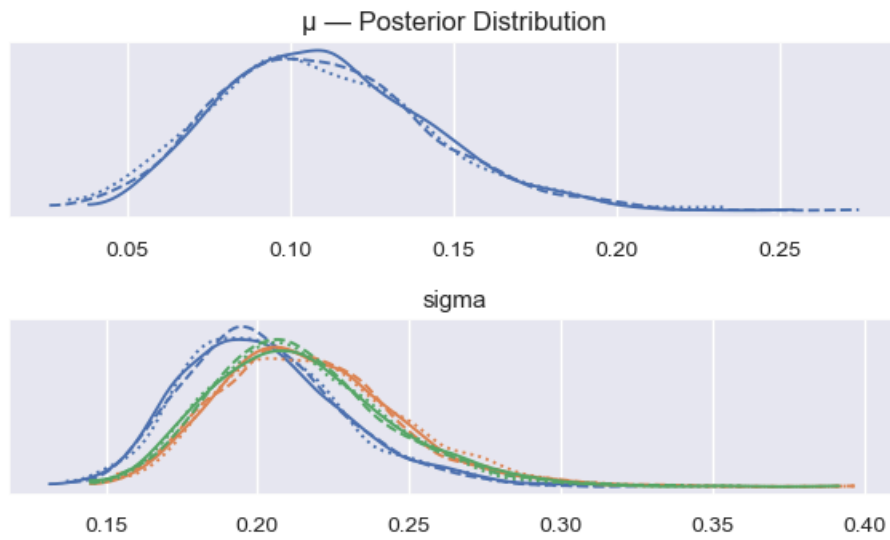
	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
mu	0.11	0.03	0.05	0.17	0.0	0.0	5678.0	3367.0	1.0

```

In [18]: axes = az.plot_trace(trace1)
axes[0, 0].set_title("μ - Posterior Distribution") # top-left (KDE)
axes[0, 1].set_title("μ - Trace Plot")             # top-right (trace)
plt.suptitle("I, R, D observed", fontsize=14, y=1.02)
plt.tight_layout()
plt.show()

```

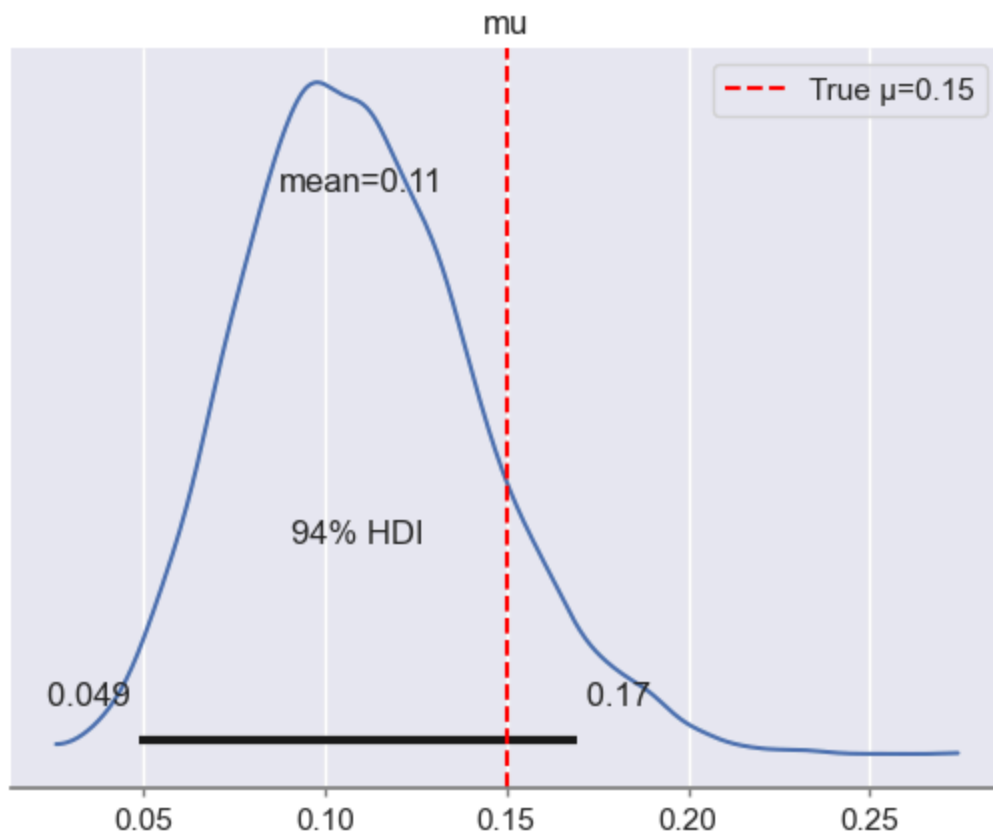
I, R, D observed



```
In [19]: az.plot_posterior(trace1, var_names=["mu"], hdi_prob=0.94)
plt.axvline(mu_true, color='red', linestyle='--', label='True μ=0.15')
plt.suptitle("Model (Full: I, R, D observed)")
plt.legend()
```

```
Out[19]: <matplotlib.legend.Legend at 0x300ee1f90>
```

Model (Full: I, R, D observed)



Step 7: Reduced model - no R observed

```
In [21]: with pm.Model() as model2:
# ODE system container
sird_model = DifferentialEquation(
    func = SIRD,
    times = np.arange(0,5*f,timestep_data), # Reverted to plain numpy array for times
    n_states = 4, #r(ecovered) and i(nfected) are states
    n_theta = 1, # beta=R0 only parameter
    t0 = 0 # start from zero
)

mu = pm.Lognormal("mu", mu=np.log(0.1), sigma=0.5)
sigma = pm.HalfCauchy("sigma", 1, shape=3)

sird_curves = sird_model(
    y0=np.array([s0, i0, r0, d0]), # Changed y0 to a NumPy array for better shape inference
    theta=[mu]
)
```

```
#R(t) is considered here as not given
S_hat = sird_curves[:, 0]
I_hat = sird_curves[:, 1]
D_hat = sird_curves[:, 3]
```

```
S_like = pm.Normal(
    "S_like",
    mu=S_hat,
    sigma=sigma[0],
    observed=S_obs
)
```

```
I_like = pm.Normal(
    "I_like",
    mu=I_hat,
    sigma=sigma[1],
    observed=I_obs
)
```

```
D_like = pm.Normal(
    "D_like",
    mu=D_hat,
    sigma=sigma[2],
    observed=D_obs
)
```

```
In [22]: with model2:
    trace2 = pm.sample(
        draws=2000,
        tune=1000,
        chains=3,
        target_accept=0.9,
        random_seed=42,
        return_inferencedata=True
    )
    print(pm.summary(trace2, var_names=["mu"]).round(2))
```

```
Initializing NUTS using jitter+adapt_diag...
Multiprocess sampling (3 chains in 3 jobs)
NUTS: [mu, sigma]
Output()
```

```
Sampling 3 chains for 1_000 tune and 2_000 draw iterations (3_000 + 6_000 draws total) took 317 seconds.
We recommend running at least 4 chains for robust computation of convergence diagnostics
```

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
mu	0.1	0.04	0.04	0.17	0.0	0.0	5762.0	3683.0	1.0

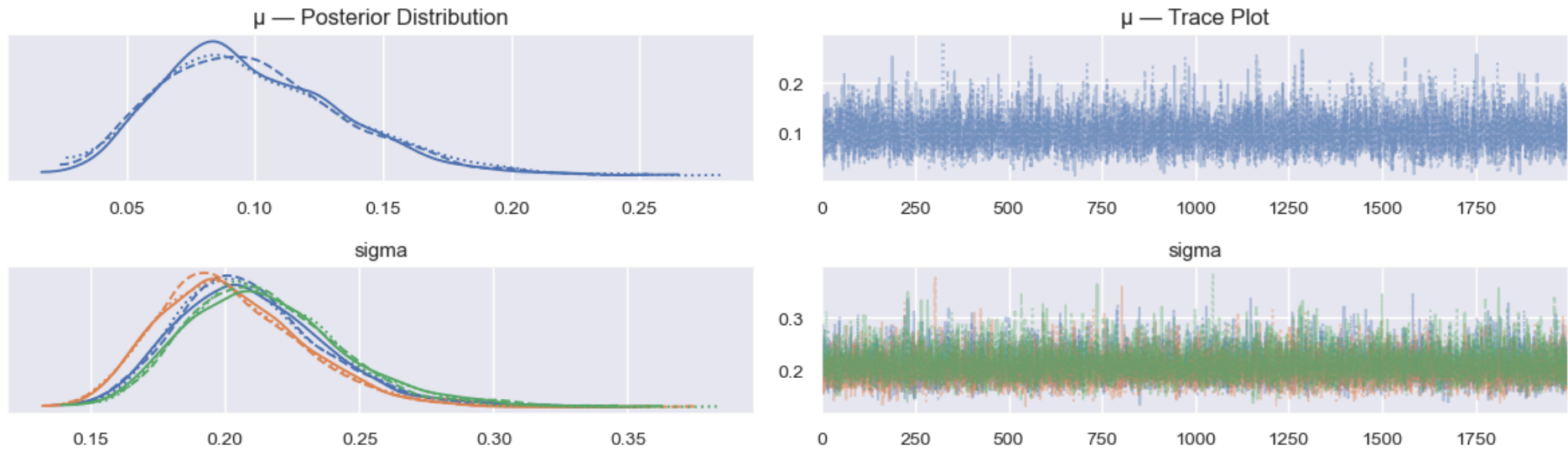
```
In [23]: axes = az.plot_trace(trace2)
    axes[0, 0].set_title("μ - Posterior Distribution")  # top-left (KDE)
```

```

axes[0, 1].set_title("μ — Trace Plot") # top-right (trace)
plt.suptitle("S, I, D observed", fontsize=14, y=1.02)
plt.tight_layout()
plt.show()

```

S, I, D observed



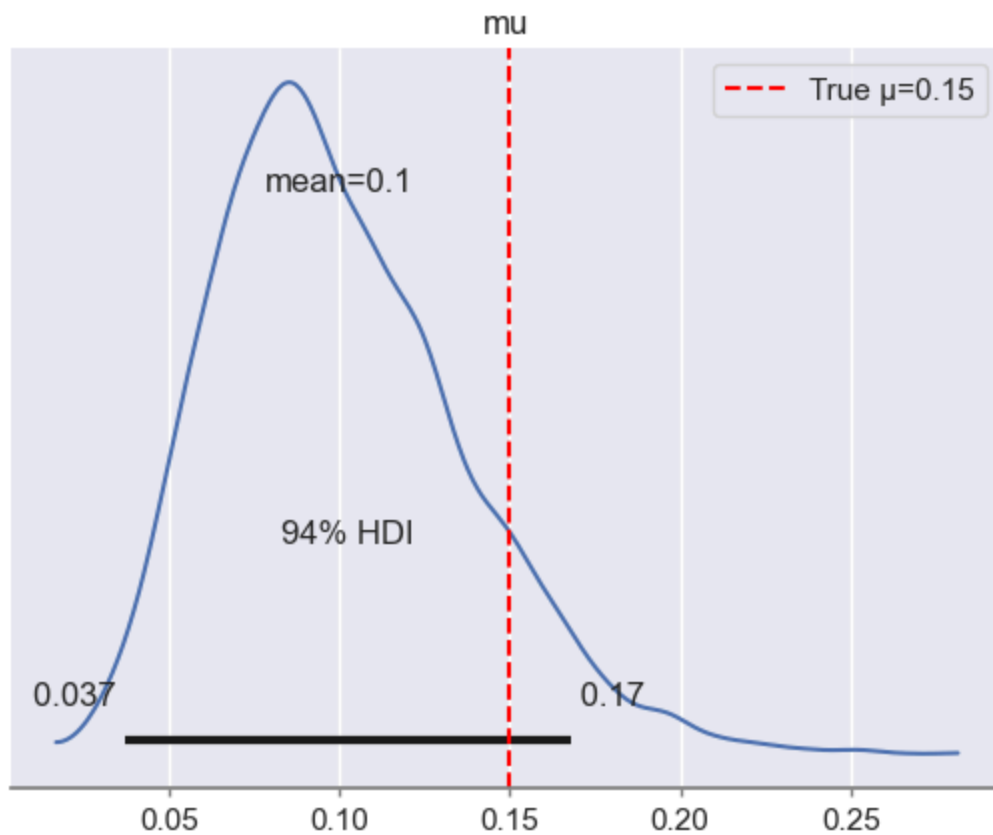
```

In [24]: az.plot_posterior(trace2, var_names=["mu"], hdi_prob=0.94)
plt.axvline(mu_true, color='red', linestyle='--', label='True μ=0.15')
plt.suptitle("Model (Full: S, I, D observed)")
plt.legend()

```

Out[24]: <matplotlib.legend.Legend at 0x3047a47d0>

Model (Full: S, I, D observed)



Step 8: Reduced model - only I and D observed

```
In [25]: with pm.Model() as model3:
# ODE system container
sird_model = DifferentialEquation(
    func = SIRD,
    times = np.arange(0,5*f,timestep_data), # Reverted to plain numpy array for times
    n_states = 4, #r(ecovered) and i(nfected) are states
    n_theta = 1, # beta=R0 only parameter
    t0 = 0 # start from zero
)

mu = pm.Lognormal("mu", mu=np.log(0.1), sigma=0.5)
sigma = pm.HalfCauchy("sigma", 1, shape=2)

sird_curves = sird_model(
    y0=np.array([s0, i0, r0, d0]), # Changed y0 to a NumPy array for better shape inference
    theta=[mu]
)
```

#both S(t) and R(t) are considered here as not given

```
I_hat = sird_curves[:, 1]
D_hat = sird_curves[:, 3]
```

```
I_like = pm.Normal(
    "I_like",
    mu=I_hat,
    sigma=sigma[0],
    observed=I_obs
)
D_like = pm.Normal(
    "D_like",
    mu=D_hat,
    sigma=sigma[1],
    observed=D_obs
)
```

In [26]: `with` model3:

```
    trace3 = pm.sample(
        draws=2000,
        tune=1000,
        chains=2,
        target_accept=0.9,
        random_seed=42,
        return_inferencedata=True
    )
print(pm.summary(trace3, var_names=["mu"]).round(2))
```

Initializing NUTS using jitter+adapt_diag...

Multiprocess sampling (2 chains in 2 jobs)

NUTS: [mu, sigma]

Output()

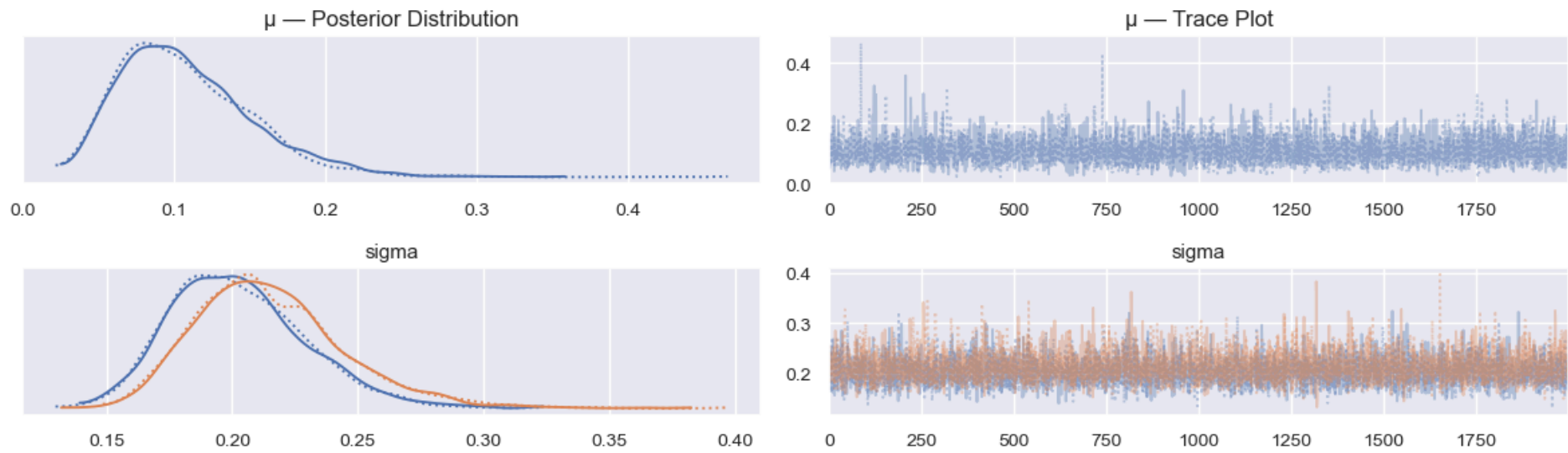
Sampling 2 chains for 1_000 tune and 2_000 draw iterations (2_000 + 4_000 draws total) took 250 seconds.

We recommend running at least 4 chains for robust computation of convergence diagnostics

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
mu	0.11	0.04	0.04	0.19	0.0	0.0	3238.0	2276.0	1.0

```
In [27]: axes = az.plot_trace(trace3)
axes[0, 0].set_title("μ - Posterior Distribution")    # top-left (KDE)
axes[0, 1].set_title("μ - Trace Plot")               # top-right (trace)
plt.suptitle("I, D observed", fontsize=14, y=1.02)
plt.tight_layout()
plt.show()
```

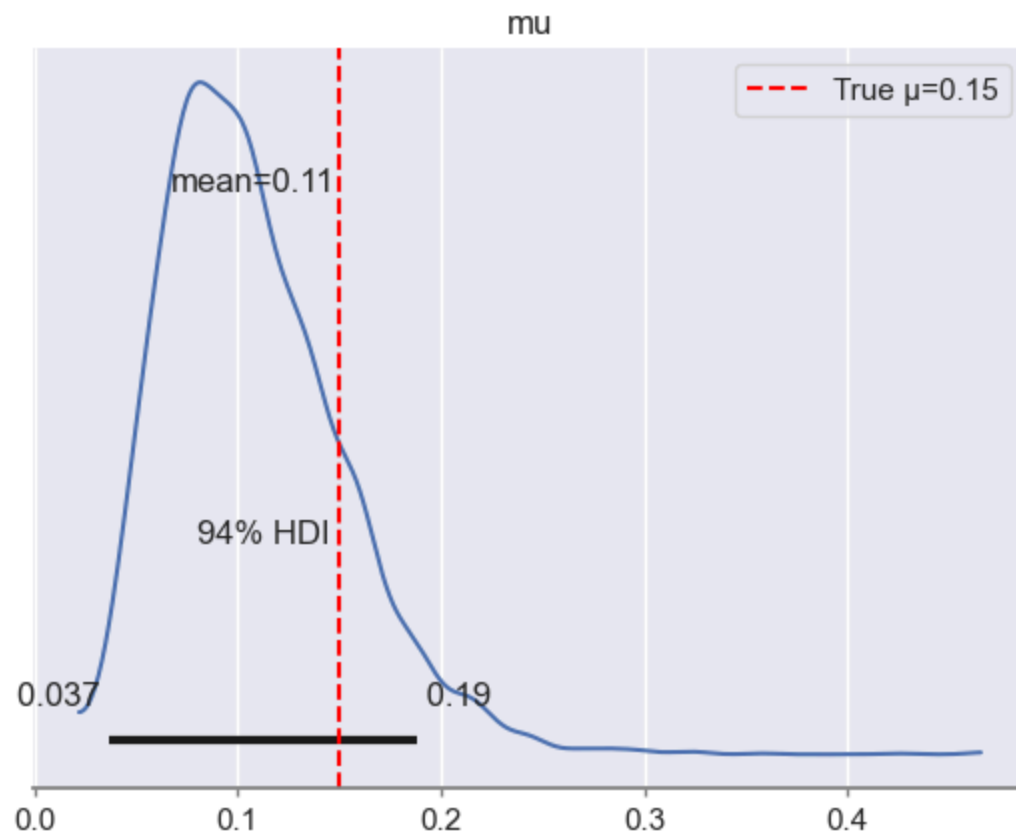
I, D observed



```
In [28]: az.plot_posterior(trace3, var_names=["mu"], hdi_prob=0.94)
plt.axvline(mu_true, color='red', linestyle='--', label='True μ=0.15')
plt.suptitle("Model (Full: I, D observed)")
plt.legend()
```

Out[28]: <matplotlib.legend.Legend at 0x305286490>

Model (Full: I, D observed)



Step 9: Comparing all models

We now compile the posterior summaries from all four models to quantitatively compare how inference quality (mean, SD, HDI width, ESS) degrades as we remove observed compartments.

```
In [ ]: summary_full = az.summary(trace, var_names=["mu"])
summary_1 = az.summary(trace1, var_names=["mu"])
summary_2 = az.summary(trace2, var_names=["mu"])
summary_3 = az.summary(trace3, var_names=["mu"])

summary_full["model"] = "S, I, R, D"
summary_1["model"] = "I, R, D"
summary_2["model"] = "S, I, D"
summary_3["model"] = "I, D"

summary_all = pd.concat([summary_full, summary_1, summary_2, summary_3])

summary_all = summary_all.reset_index()[[
```

```

"model", "mean", "sd", "hdi_3%", "hdi_97%", "ess_bulk", "r_hat"
]]

print(summary_all)

```

	model	mean	sd	hdi_3%	hdi_97%	ess_bulk	r_hat
0	S, I, R, D	0.124	0.030	0.069	0.182	8040.0	1.0
1	I, R, D	0.110	0.033	0.049	0.170	5678.0	1.0
2	S, I, D	0.100	0.037	0.037	0.168	5762.0	1.0
3	I, D	0.108	0.044	0.037	0.188	3238.0	1.0

- As the number of observed compartments decreases, the **posterior mean shifts from 0.124** (full model) to values around **0.10–0.11** in the reduced models. Meanwhile, the posterior standard deviation increases from **0.030 to 0.044**, indicating growing uncertainty.
- The width of the **94%** HDI expands accordingly, while the effective sample size decreases, reflecting reduced sampling efficiency. However, all models yield posterior intervals that include the true value **$\mu=0.15$** .

Key Observations and Takeaways

1. The SIRD model is identifiable from noisy data.

- Even with noise $\sigma = 0.2$ added to all compartments, MCMC successfully includes the true value $\mu \approx 0.15$ within the posterior credible intervals in all four experiments, indicating successful inference.
- The Gelman–Rubin diagnostic $\hat{r} = 1.0$ in every run confirms reliable convergence.

2. μ can be inferred even without $S(t)$ or $R(t)$.

- Dropping $S(t)$, dropping $R(t)$, or dropping both still yields posterior distributions that cover the true μ . Since

$$D(t) = \int \mu \cdot I(t) dt,$$

the signal for μ is primarily carried by the D and I compartments.

- The S and R compartments contribute indirectly by constraining the overall epidemic trajectory.

3. Uncertainty grows as observability is reduced.

- The ordered increase in posterior standard deviation

$$0.030 \rightarrow 0.033 \rightarrow 0.037 \rightarrow 0.044$$

and the overall decrease in effective sample size

$$8040 \rightarrow 5678 \rightarrow 5762 \rightarrow 3238$$

are physically meaningful.

- With fewer observed compartments, the likelihood becomes flatter, causing the sampler to explore a broader parameter region and mix less efficiently.

4. **The LogNormal prior on μ is appropriate.**

- The prior with log-mean $\log(0.1)$ and standard deviation $\sigma = 0.5$ is broad enough not to dominate the posterior while correctly enforcing the physical constraint $\mu > 0$.
- The posterior mean shifts away from the prior center toward the true value, confirming that the data are informative.

5. **MCMC diagnostics are healthy throughout.**

- ESS values range from approximately 3000 to 8000, well above the common rule-of-thumb threshold of approximately 400 per chain.
- No divergences were flagged in any run, and all models achieved $\hat{r} = 1.0$, indicating stable sampling and reliable posterior exploration.